
Generalized Social Marginal Welfare Weights for Optimal Tax Theory

Author(s): Emmanuel Saez and Stefanie Stantcheva

Source: *The American Economic Review*, JANUARY 2016, Vol. 106, No. 1 (JANUARY 2016), pp. 24-45

Published by: American Economic Association

Stable URL: <https://www.jstor.org/stable/43821396>

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



JSTOR

is collaborating with JSTOR to digitize, preserve and extend access to *The American Economic Review*

Generalized Social Marginal Welfare Weights for Optimal Tax Theory[†]

By EMMANUEL SAEZ AND STEFANIE STANTCHEVA*

This paper proposes to evaluate tax reforms by aggregating money metric losses and gains of different individuals using “generalized social marginal welfare weights.” Optimum tax formulas take the same form as standard welfarist tax formulas by simply substituting standard marginal social welfare weights with those generalized weights. Weights directly capture society’s concerns for fairness without being necessarily tied to individual utilities. Suitable weights can help reconcile discrepancies between the welfarist approach and actual tax practice, as well as unify in an operational way the most prominent alternatives to utilitarianism such as Libertarianism, equality of opportunity, or poverty alleviation. (JEL D60, D63, H21, H23, I38)

This paper proposes a novel approach to optimal tax theory using generalized social marginal welfare weights. In our approach, presented in Section I, there is no social welfare objective primitive that the government maximizes. Instead, our primitives are *generalized social marginal welfare weights* which represent the value that society puts on providing an additional dollar of consumption to any given individual.¹ These weights directly reflect society’s concerns for fairness. Equipped with such weights, we can evaluate small budget neutral tax reforms by simply aggregating money metric utility gains and losses across individuals using the weights. If the net aggregate gain is positive, the reform is desirable (and conversely). We define a tax system as locally optimal if no small reform is desirable.² This has four implications.

First, optimal tax formulas in our theory take the same form as optimal tax formulas in the standard approach by simply substituting standard social welfare weights with our generalized weights. Hence, our theory remains as tractable as the standard

* Saez: University of California, 530 Evans Hall #3880, Berkeley, CA 94720 (e-mail: saez@econ.berkeley.edu). Stantcheva: Society of Fellows, Harvard University, Cambridge, MA 02138 (e-mail: sstantcheva@fas.harvard.edu). We thank Steve Coate, Florian Ederer, Marc Fleurbaey, Bas Jacobs, Louis Kaplow, Henrik Kleven, Etienne Lehmann, Ben Lockwood, Thomas Piketty, Maxim Troshkin, Aleh Tsyvinski, Matthew Weinzierl, Nicolas Werquin, three anonymous referees, and numerous conference participants for useful discussions and comments. We acknowledge financial support from NSF grant SES-1156240, the MacArthur Foundation, and the Center for Equitable Growth at UC Berkeley. The authors declare that they have no relevant or material financial interests that relate to the research described in this paper.

[†] Go to <http://dx.doi.org/10.1257/aer.20141362> to visit the article page for additional materials and author disclosure statement(s).

¹ We take society’s preferences as given and do not analyze how they could arise through the political process.

² With the tax reform approach, we can evaluate beneficial reforms even starting from nonoptimal tax systems.

approach and can easily be operationalized for any specification of generalized welfare weights. Second, our theory nests the standard welfarist approach where the government maximizes a social welfare function that depends solely on individual utilities if we define generalized social welfare weights to be equal to the standard marginal social welfare weights.³ Third, if the weights are nonnegative, then our theory respects the Pareto principle in the sense that, around the local optimum, there is no Pareto improving small reform. Fourth, our theory is by nature local as we can only evaluate small changes around a given tax/transfer system, but we have no way of systematically comparing two local optima. In most applications, however, the local optimum will be unique.

The key advantage of our approach is that weights can be defined very generally allowing us to capture a broader set of concepts of justice than the standard approach. Weights can depend on individual and aggregate characteristics, some of which are endogenous to the tax and transfer system. The characteristics which enter the welfare weights determine the dimensions along which society considers redistribution to be fair. These characteristics could be part of individuals' utilities (in the welfarist spirit). Importantly, though, social welfare weights can also depend on individual or aggregate characteristics which do not enter individuals' utilities. Conversely, the welfare weights can omit some characteristics which enter individuals' utility functions, but for which society does not deem it fair to compensate individuals.

In Section II, we contrast our approach with the standard approach through several examples. First, we show that making generalized social marginal weights depend not only on net disposable income but also on net taxes paid produces a nondegenerate optimal tax theory even absent behavioral responses. Second, generalized social weights can depend on what individuals would have done absent taxes and transfers. Hence, we can capture the idea that society dislikes marginal transfers toward "freeloaders" who would work absent means-tested transfers. Third, our approach can capture horizontal equity concerns. A reasonable criterion is that introducing horizontal inequities is acceptable only if it benefits the group discriminated against. This dramatically limits the scope for using non-income based tags.

In Section III, we show how the most prominent alternatives to welfarism can be re-cast within our theory, i.e., we can derive the generalized social welfare weights implied by those alternative theories. First, the equality of opportunity principle developed by Roemer (1998) and Roemer et al. (2003) concentrates weights uniformly on those coming from a disadvantaged background as, conditional on earnings, they have more merit (have worked harder) than those coming from an advantaged background. As the likelihood of coming from a disadvantaged background decreases with income, social weights decrease with income for a reason completely orthogonal to the decreasing marginal utility of income in utilitarianism. It also provides a rationale for less progressive taxes when there is high social or intergenerational mobility. Second, a poverty alleviation objective that respects the

³To be precise, our theory nests the first order approach of the standard social welfare function maximization. As is well known (see, e.g., Piketty and Saez 2013a for a survey), the optimum of the standard approach is also such that no small reform can increase social welfare and the welfare effect of the reform is the aggregate of gains and losses using the standard social marginal welfare weights. One advantage of the standard approach relative to ours is that local optima (if they are multiple) can be ranked using the primitive social welfare function.

Pareto principle can be captured by social welfare weights concentrated on those below the poverty threshold.⁴

All formal proofs, several theoretical extensions, as well as results from a simple online survey designed to elicit social preferences of subjects are presented in the online Appendix.

I. Presentation of Our Approach

Individual Utilities and Taxes.—We present our approach using income taxation.⁵ Consider a population with a continuum of individuals indexed by i . Population size is normalized to one. Individual i derives utility from consumption c_i and incurs disutility from earning income $z_i \geq 0$, with a utility representation

$$u_i = u(c_i - v(z_i; x_i^u, x_i^b)),$$

where x_i^u and x_i^b are sets of characteristics. The functions u and v are common to all individuals. u is increasing and concave and v is increasing and convex in z . The quasilinear functional forms rule out income effects on earnings which greatly simplifies optimal tax formulas (see below). u_i is a cardinal utility representation for individual i as viewed by the Planner (see below). x^u are characteristics that exclusively enter the utility function, while x^b will be characteristics that also affect the generalized social welfare weights introduced below.⁶ Two important individual characteristics are considered throughout the paper. First, we consider a person's productivity per unit of effort $w_i \equiv z_i/l_i$ where l_i is labor supply. w_i is distributed in the population with a density $f(w)$ on $[w_{\min}, w_{\max}]$. Second, we consider the cost of work θ_i that affects the disutility from producing any unit of effort, distributed according to a distribution $p(\theta)$ on $[\theta_{\min}, \theta_{\max}]$. For instance, the disutility from labor could take the form $\theta_i \tilde{v}(z_i/w_i)$, so that any unit of effort is more costly for high θ_i agents.

The government sets an income tax $T(z)$ as a function of earnings only so that $c_i = z_i - T(z_i)$.⁷ Individual i chooses z_i to maximize $u(z_i - T(z_i) - v(z_i; x_i^u, x_i^b))$.

Generalized Social Marginal Welfare Weights.—We propose a novel theory of taxation that starts from the social welfare weights. For any individual, we define a *generalized social marginal welfare weight* g_i which measures how much society values the marginal consumption of individual i . We will assess the welfare gains from any reform by weighting the money metric welfare gains or losses to each individual using these weights and characterize optimal tax systems as systems around which no small reform can yield a welfare gain.

⁴In online Appendix B, we consider additional alternatives to welfarism, such as Libertarianism, Rawlsianism, or the fair income tax of Fleurbaey and Maniquet (2011) and show how they map into social weights.

⁵The same approach can be applied to other forms of taxation or transfer policies.

⁶It is possible to consider a more general utility with $u_i = u(x_i^f \cdot c_i - v(z_i; x_i^u, x_i^b))$ where x_i^f would be a shifter parameter for the marginal utility from consumption. For simplicity, we abstract from this heterogeneity as none of our examples require it.

⁷Section IIC considers tax systems that can also depend on other observable individual characteristics ("tags").

DEFINITION 1: *The generalized social marginal welfare weight on individual i is $g_i = g(c_i, z_i; x_i^s, x_i^b)$ where g is a function, x_i^s is a set of characteristics which only affect the social welfare weight, while x_i^b is a set of characteristics which also affect utility.*

Naturally, the generalized weights are only defined up to a multiplicative constant as they measure only the *relative* value of consumption of individual i . Importantly, they are allowed to depend on individual characteristics x_i^s and x_i^b . These characteristics may be unobservable to the government or it may be impossible or unacceptable to condition the tax system on them. They nevertheless enter the social welfare weights because they affect how deserving a person is deemed by society. For instance, x_i^s might include family background (see our treatment of “Equality of Opportunity” in Section IIIA), a characteristic that typically does not affect one’s taxes directly but affects perceptions of deservedness.

As depicted in Figure 1, there is an important conceptual distinction between the sets of characteristics x^b , x^u , and x^s . Characteristics which enter the social welfare weight are dimensions that society considers potentially *fair to redistribute across* and to compensate for. For instance, if the disutility of work θ_i is due mostly to differences in health status or disability, then it might be fair to include it in the social welfare weight (see the “Freeloaders” example in Section IIB). Conversely, if differences in disutility of work are mostly based on preferences for leisure, then they might not enter the social welfare weight (see the “Fair Income Tax” example in online Appendix B5). These value judgments are directly embodied in the specification of the social welfare weights.

We keep individual preferences standard and individuals’ utility maximization intact. Although it is possible to modify individual preferences to directly incorporate justice and fairness criteria (see Alesina and Angeletos 2005; Fehr and Schmidt 1999), that would still leave open the question of how to aggregate individual preferences.

Tax Reform Analysis.—We now turn to defining desirable tax reforms around $T(z)$ and an optimal tax criterion. For a given ε , we define the perturbed tax system in the direction ΔT as the tax system $z \rightarrow T(z) + \varepsilon \Delta T(z)$. We will derive the behavioral responses, revenue changes, and social welfare changes when $\varepsilon \rightarrow 0$, i.e., according to the Gâteaux differential.⁸ In the main text, we consider only the first order approach because of its simplicity and its applicability to all cases we consider later on. We show in online Appendix A1 how to formally consider second order conditions and a fully rigorous local optimization approach.

Denote by $z_i(T)$ the earning choice of agent i under tax system T . The behavioral response induced by the tax reform of system T in the direction ΔT is $\Delta z_i \equiv \lim_{\varepsilon \rightarrow 0} \frac{z_i(T + \varepsilon \Delta T) - z_i(T)}{\varepsilon}$. Using the standard envelope argument from the individual’s optimization of z_i , the small tax reform $\varepsilon \Delta T$ mechanically changes individual disposable income by $-\varepsilon \Delta T(z_i)$, but the induced

⁸ Effectively, this amounts to considering first order effects in ε . We always consider interior tax systems so that if a small reform in the direction ΔT is feasible, a small reform in direction $-\Delta T$ is also feasible.

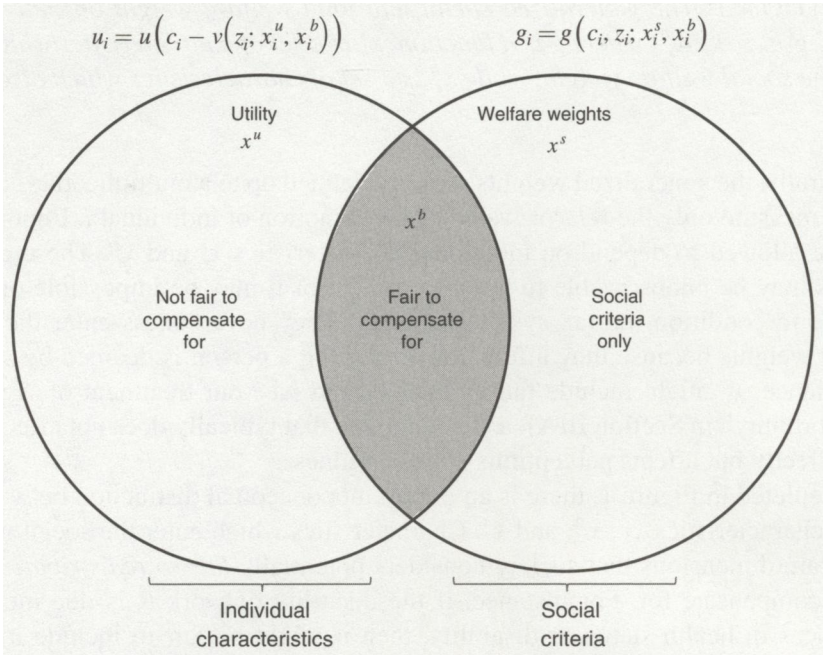


FIGURE 1. GENERALIZED SOCIAL WELFARE WEIGHTS APPROACH

Notes: This figure depicts the three sets of individual characteristics x^b , x^u , and x^s . Characteristics x^u enter solely the utility function (i.e., they affect individual utilities and choices). Characteristics x^s enter solely the generalized social welfare weights (i.e., they affect how society values marginal transfers to each individual). Characteristics x^b enter both the utility function and social weights.

behavioral response in earnings $\varepsilon \Delta z_i$ has no first-order effect on utility. We denote by $U_i(T)$ the indirect utility of agent i under T , the utility first order change is $\Delta U_i \equiv \lim_{\varepsilon \rightarrow 0} \frac{U_i(T + \varepsilon \Delta T) - U_i(T)}{\varepsilon} = -u_{c_i} \cdot \Delta T(z_i)$.⁹ Therefore, $\Delta T(z_i)$ measures the money-metric first order welfare impact of the tax reform in direction ΔT on individual i .¹⁰ Hence, the first order effect on welfare is $-\int_i g_i \Delta T(z_i) \, di$ where di denotes the integral measure over all agents i in the population.

Next, we define a small tax reform as being budget neutral if it has no first order effect on net tax revenue (i.e., the Gâteaux differential of tax revenue is zero).

DEFINITION 2 (Budget Neutral Tax Reform): A small tax reform in direction $\Delta T(z)$ is budget neutral if and only if $\lim_{\varepsilon \rightarrow 0} \left\{ \int_i [T(z_i(T + \varepsilon \Delta T)) + \varepsilon \Delta T(z_i(T + \varepsilon \Delta T))] \, di - \int_i T(z_i(T)) \, di \right\} / \varepsilon = 0$.

⁹To see this, denoting by $u_i(c, z)$ individual i utility function, we have the following first order expansion: $U_i(T + \varepsilon \Delta T) = u_i(z_i(T + \varepsilon \Delta T) - T(z_i(T + \varepsilon \Delta T)) - \varepsilon \Delta T(z_i(T + \varepsilon \Delta T)), z_i(T + \varepsilon \Delta T)) = U_i(T) - u_{c_i} \cdot \varepsilon \Delta T(z_i) + [u_{c_i} \cdot (1 - T'(z_i)) + u_{z_i}] \cdot \varepsilon \Delta z_i + o(\varepsilon) = -u_{c_i} \cdot \varepsilon \Delta T(z_i) + o(\varepsilon)$ as $u_{c_i} \cdot (1 - T'(z_i)) + u_{z_i} = 0$ by individual maximization.

¹⁰The incentive constraints are reflected in the behavioral response to the tax change. There are revenue effects from the behavioral responses. The welfare effect calculus requires that there are no externalities of individuals' actions and that the social welfare function is not paternalistic (i.e., respects individuals' optimization).

In Definition 2, $\varepsilon \rightarrow 0$ both from the right ($\varepsilon > 0$) and left ($\varepsilon < 0$). Hence, if a small reform in direction $\Delta T(z)$ is budget neutral then a reform in direction $-\Delta T(z)$ is also budget neutral.

Next, we define a tax reform as being desirable if its first order effect on welfare is positive.

DEFINITION 3 (Tax Reform Desirability Criterion): A small budget neutral tax reform around $T(z)$ in direction $\Delta T(z)$ is desirable if and only if $\int_i g_i \Delta T(z_i) di < 0$, with g_i the generalized social marginal welfare weight on individual i evaluated at $(c_i = z_i - T(z_i), z_i, x_i^s, x_i^b)$.

Last, we can define a locally optimal tax system as a tax system around which no budget neutral tax reform can improve welfare. Note that if $\int_i g_i \Delta T(z_i) di > 0$ then $\int_i g_i \cdot (-\Delta T(z_i)) di < 0$. This immediately gives necessary conditions for a local optimum:¹¹

PROPOSITION 1 (Local Optimal Tax Criterion): If a tax system $T(z)$ is a local optimum, then for any small budget neutral tax reform in direction $\Delta T(z)$, $\int_i g_i \Delta T(z_i) di = 0$, with g_i the generalized social marginal welfare weight on individual i evaluated at $(z_i - T(z_i), z_i, x_i^s, x_i^b)$.

As in the standard theory (see, e.g., Kleven and Kreiner 2006; Chetty 2009; Hendren 2013), the tax reform approach requires knowing only the weights g_i and the behavioral responses around the current system while the optimal tax criterion requires knowing the weights g_i and the behavioral responses *at the optimum*. Hence, the tax reform is more readily applicable.

From Individual Weights to Applicable Weights in the Tax Formula.—The weights on each individual g_i are not immediately applicable. As noted, they can depend on unobservable characteristics or elements that the tax system cannot condition upon. They merely embody society's judgment of fairness, without taking into account observability or feasibility. To apply the weights to the evaluation of tax systems, the individual weights need to be "aggregated" up to only those characteristics that the tax system can be conditioned on. For a tax system purely based on income, $T(z)$, the weights need to be aggregated at each income level z .

Local Optimal Tax Formulas.—With Proposition 1, we can obtain an optimal tax formula that is particularly simple in the case with no income effects on labor supply that we consider here (Diamond 1998). The proof is the same as in the standard approach (see Piketty and Saez 2013a) and is presented in online Appendix A2.¹²

¹¹ We discuss second order conditions in online Appendix A1. The extra complexity required is not needed in the applications we consider later on and hence relegated to the online Appendix.

¹² Using small tax reforms around the optimum, Saez (2001) informally derived an optimal income tax formula that generalizes the formulas of Mirrlees (1971) to situations with heterogeneous populations, where individuals differ not only in skills but also possibly in preferences. Jacquet and Lehmann (2015) provide a fully rigorous proof of the formula with heterogeneous populations and give conditions under which it applies. We always assume here that these conditions hold.

Let $H(z)$ denote the cumulative earnings distribution function and $h(z)$ the earnings density. Both naturally depend on the tax system.

DEFINITION 4: Let $\bar{G}(z)$ be the (relative) average social marginal welfare weight for individuals who earn more than z :

$$(1) \quad \bar{G}(z) \equiv \frac{\int_{\{i: z_i \geq z\}} g_i \, di}{\text{Prob}(z_i \geq z) \cdot \int_i g_i \, di}.$$

Let $\bar{g}(z)$ be the corresponding average social marginal welfare weight at earnings level z , with $\bar{G}(z)[1 - H(z)] = \int_z^\infty \bar{g}(z') \, dH(z')$, or, equivalently $\bar{g}(z) = -\frac{1}{h(z)} \frac{d(\bar{G}(z) \cdot [1 - H(z)])}{dz}$.

From Definition 4, we have $\bar{G}(0) = \int_0^\infty \bar{g}(z) \, dH(z) = 1$, so that the weights $\bar{g}(z)$ average to one.

PROPOSITION 2: The optimal marginal tax rate at income level z satisfies the formula

$$(2) \quad T'(z) = \frac{1 - \bar{G}(z)}{1 - \bar{G}(z) + \alpha(z) \cdot e(z)},$$

with $e(z)$ the average elasticity of earnings z_i with respect to the retention rate $1 - T'$ for individuals earning $z_i = z$, $\alpha(z)$ the local Pareto parameter defined as $zh(z)/[1 - H(z)]$.¹³

The optimal tax formula looks exactly as in Saez (2001), replacing the standard weights by the generalized social marginal welfare weights, averaged at each income level. Accordingly, some standard results would hold under the same conditions. For instance, $T'(z) \geq 0$ if average social marginal welfare weights $\bar{g}(z)$ are decreasing in income.¹⁴ The optimal marginal tax rate at the very top is zero if the income distribution is bounded.

We can similarly express the optimal linear tax rate τ as a function of the welfare weights:¹⁵

PROPOSITION 3: The optimal linear income tax rate satisfies the formula

$$(3) \quad \tau = \frac{1 - \bar{g}}{1 - \bar{g} + e} \quad \text{with} \quad \bar{g} \equiv \frac{\int_i g_i \cdot z_i \, di}{\int_i g_i \, di \cdot \int_i z_i \, di}$$

¹³The local Pareto parameter $\alpha(z)$ is constant and equal to the Pareto parameter for Paretian distributions. To be precise, when defining $\alpha(z)$, $h(z)$ is defined as the virtual density that would hold at z if the income tax system were linearized at z . See Saez (2001) and Piketty and Saez (2013a) for details.

¹⁴This condition is satisfied with standard utilitarian weights. See online Appendix A2 for details on these results.

¹⁵See again online Appendix A2 for a short derivation and Piketty and Saez (2013a) for details.

and e the elasticity of aggregate income $\int_i z_i di$ with respect to the retention rate $(1 - \tau)$.

$\bar{g}$ can be interpreted as the average g_i weighted by income z_i (relative to the population average g_i) or symmetrically as the average z_i weighted by g_i (relative to $\int_i z_i di$).

As in standard optimal tax theory, formulas (2) and (3) hold at an optimum but are not necessarily sufficient. Importantly, there may be several tax systems that can satisfy the formulas. Our theory is fundamentally local and hence cannot rank optima when they are multiple. This is a disadvantage relative to any approach that maximizes a social objective and hence provides a complete ordering of all tax systems.

Advantages of the Generalized Approach.—First, our approach allows us to re-use existing tax formulas such as equations (2) and (3) once they are written in terms of the welfare weights at each income level. Hence, it also nests the standard approach. Second, it can easily ensure that any tax optimum is (locally) constrained Pareto efficient as long as the generalized weights g_i are all nonnegative. In that case, our theory respects individual preferences. Third, and most importantly, as we will outline throughout the paper, our approach grants great flexibility in the choice of the welfare weights g_i . The fact that social welfare weights can depend on characteristics outside of individuals' utilities, as well as ignore characteristics from individuals' utilities—as illustrated in Figure 1—allows us to incorporate elements that matter in actual tax policy debates but that cannot be captured with the standard welfarist approach.¹⁶

Implicit Pareto Weights.—As is well known, under standard regularity conditions, any second-best allocation can be obtained maximizing a linear social welfare function $SWF = \int_i (\omega_i \cdot u_i) di$ with suitable Pareto weights $\omega_i \geq 0$. The following proposition extends this result to our setting.

PROPOSITION 4: *For any nonnegative generalized weights function and a local optimum $T(z)$, there exist Pareto weights $\omega_i \geq 0$ such that $T(z)$ satisfies the first order condition formula (2) of the maximization of $SWF = \int_i (\omega_i \cdot u_i) di$. The Pareto weights are such that $\omega_i = g_i / u_{c_i} \geq 0$ where g_i and u_{c_i} are evaluated at the optimum allocation.*

Hence, our approach can be reverse-engineered to obtain a set of Pareto weights ω_i and a corresponding standard social welfare function $\int_i (\omega_i \cdot u_i) di$. Importantly, because g_i / u_{c_i} is evaluated at the optimum allocation, it is taken as fixed in the maximization of SWF . In practice, as we shall see, it is impossible to posit the correct weights ω_i without *first* having solved for the optimum using our approach that starts with the social marginal weights g_i . In addition, it is typically impossible to translate the generalized social welfare weights into Pareto weights that are fixed and hence cannot be modified with changes in the environment. A social welfare

¹⁶The tax reform approach, which is *per se* not specific to the generalized welfare weights, has the advantage that it can evaluate reforms even starting from nonoptimal systems.

function with endogenous Pareto weights such as $\int_i \omega_i(T) u_i \, di$ would not deliver the same solution as our approach and could lead to Pareto dominated outcomes.¹⁷

II. Enriching the Welfarist Approach

In this section, we show how the use of suitable generalized social marginal welfare weights can enrich the welfarist approach by reconciling some of the discrepancies between the welfarist approach (and particularly the widely used utilitarian case) and actual tax practice. Table 1 summarizes these results by contrasting actual tax practice (column 1), the standard welfarist approach (column 2), and our generalized social marginal welfare weights approach (column 3) in various situations. In each situation, column 3 indicates what property of social marginal welfare weights is required to make this approach fit with actual tax policy practice.

A. Optimal Tax Theory with Fixed Incomes

We start with the simple case in which pretax incomes are completely inelastic to taxes and transfers. This puts the focus solely on redistributive issues. It is a useful illustration of our approach, especially as contrasted with the standard welfarist approach. We specialize our general framework with a disutility of work $v(z; z_i^0) = 0$ if $z \leq z_i^0$ and $v(z; z_i^0) = \infty$ if $z > z_i^0$. Thus, z_i^0 is an exogenous characteristic of individual i , contained in x_i^u , and choosing $z_i = z_i^0$ is always optimal for individual i , so that the distribution of incomes is exogenous to the tax system. In equilibrium, utility is $u_i = u(c_i)$. We review first the standard utilitarian setting.

Standard Utilitarian Approach.—The government chooses $T(z)$ to maximize $\int_i u(z_i - T(z_i)) \, di$ subject to the resource constraint $\int_i T(z_i) \, di \geq 0$ (with multiplier p). A point-wise maximization with respect to $T(z)$ yields $u'(z - T(z)) = p$ so that $c = z - T(z)$ is constant across z . Hence, utilitarianism with inelastic earnings and a concave individual utility function, homogeneous across individuals, leads to complete redistribution of incomes. The government taxes 100 percent of earnings and redistributes income equally across individuals (Edgeworth 1897).

This simple case highlights three drawbacks of utilitarianism. First, complete redistribution seems too strong a result. In reality, even absent behavioral responses, many and perhaps even most people would still object to full taxation on the grounds that it is unfair to fully confiscate individual incomes. Second, the outcome is extremely sensitive to the specification of individual utilities, as linear utility calls for no taxes at all, while introducing just a bit of concavity leads to complete redistribution. Third, the utilitarian approach cannot handle well heterogeneity in individual utility functions, a problem known as interpersonal utility comparisons. To see this, suppose that utilities are heterogeneous of the form $u_i = x_i \cdot u(c_i)$. The optimum is such that $x_i \cdot u'(c_i)$ are equal for all i . Hence, consumption should be

¹⁷ A closely related point has also been made by Fleurbaey and Maniquet (2015, p. 28) who show that their fair income tax social objective maximization can also be obtained as the maximization of a weighted sum of utilities but the weights “would have to be computed for each new problem, that is, as a function of the set of allocations among which the choice has to be made, [...], and could only be ascertained after the optimal allocation is identified.”

TABLE 1—GENERALIZED SOCIAL MARGINAL WELFARE WEIGHTS

	Actual practice (1)	Standard welfare criterion (2)	Generalized social marginal welfare weights (3)
Pareto efficiency	Desirable	Yes	Yes (local) if g nonnegative
Optimal taxes with fixed incomes	Nondegenerate	Degenerate (Full redistribution desirable)	g must depend on taxes paid in addition to consumption
Freeloaders	Important	Cannot capture counterfactuals	Can be captured if g depends on counterfactual (whether work absent transfers)
Tagging	Used minimally	Highly desirable	Can justify nondesirability if g depends on horizontal equity
Poverty alleviation	Relevant criterion	Direct poverty minimization leads to Pareto inefficiency	Can maintain Pareto efficiency

Notes: This table contrasts actual practice (column 1), the standard welfarist approach (column 2), and our generalized social marginal welfare weights approach (column 3) in various situations listed on the left-hand side of the table. In each situation, column 3 indicates what property of social marginal welfare weights (denoted by g) is required to make this approach fit with actual tax policy practice.

higher for individuals with higher x_i , i.e., individuals more able to enjoy consumption. In reality, society would be reluctant to redistribute based on preferences, as confirmed by our online survey in online Appendix C.¹⁸

Generalized Social Marginal Welfare Weights.—The simplest way to illustrate the power of our approach with fixed incomes is to use the generalized weights from Definition 1 without using any additional characteristics ($x_i^u = z_i^0$, for all i , and x^b and x^s are empty).

DEFINITION 5 (Simple Generalized Weights): Let $g_i = g(c_i, z_i) = \tilde{g}(c_i, z_i - c_i)$ with $\tilde{g}_c \equiv \frac{\partial \tilde{g}}{\partial c} \leq 0$, $\tilde{g}_{z-c} \equiv \frac{\partial \tilde{g}}{\partial (z - c)} \geq 0$. There are two polar cases of interest:

- (i) *Utilitarian weights:* $g_i = g(c_i, z_i) = \tilde{g}(c_i)$ for all z_i , with $\tilde{g}(\cdot)$ decreasing.
- (ii) *Libertarian weights:* $g_i = g(c_i, z_i) = \tilde{g}(z_i - c_i)$ with $\tilde{g}(\cdot)$ increasing.

Weights depend not only negatively on c but also positively on net taxes paid $z - c$. $\tilde{g}_c \leq 0$ can reflect decreasing marginal utility of consumption as under utilitarianism or more generally the old notion of “ability to pay” (those with higher disposable income can afford paying \$1 extra in taxes more easily). $\tilde{g}_{z-c} \geq 0$ reflects the fact that taxpayers contribute more to society and hence are more deserving. Alternatively, individuals are entitled to their income and hence become more deserving as the government taxes away their income.^{19, 20}

¹⁸Redistribution based on marginal utility is socially acceptable if there are objective reasons for a person to have higher needs, such as having a medical condition requiring high expenses, or a large family with many dependents.

¹⁹This is akin to an “Equal Sacrifice” principle in money metric utility terms. Weinzierl (2014) also incorporates a libertarian element in an optimal tax model by considering a mixed objective.

²⁰We assume away government funded public goods in our setup for simplicity.

The optimal tax system, according to Proposition 1 is such that no reform can increase social welfare at the margin, where transfers are evaluated using the g weights. Since, $T(z) = z - c$, $\tilde{g}(c, z - c) = \tilde{g}(z - T(z), T(z))$. With no behavioral responses, the optimal rule is simple: social welfare weights $\tilde{g}(z - T(z), T(z))$ need to be equalized across all incomes z . Intuitively, if we had nonequalized weights with $\tilde{g}(z_1 - T(z_1), z_1) > \tilde{g}(z_2 - T(z_2), z_2)$, transferring a dollar from those earning z_2 toward those earning z_1 (by adjusting $T(z_1)$ and $T(z_2)$ correspondingly and in a budget balanced manner) would be desirable (the formal proof is in online Appendix A4).

PROPOSITION 5: *The optimal tax schedule with no behavioral responses is such that*

$$(4) \quad T'(z) = \frac{1}{1 - \tilde{g}_{z-c}/\tilde{g}_c} \quad \text{so that} \quad 0 \leq T'(z) \leq 1.$$

COROLLARY 1: *In the standard utilitarian case, $T'(z) \equiv 1$. In the libertarian case, $T'(z) \equiv 0$.*

We present in online Appendix C results from a survey asking subjects to rank taxpayers with various incomes and tax burdens in terms of deservedness of a tax break. Respondents put weight on both disposable income and gross income, showing that social preferences are in between the polar utilitarian and libertarian cases. Such data can be used to recover social preferences $g(c, z)$. For instance, the specification $g(c, z) = \tilde{g}(c - \alpha(z - c)) = \tilde{g}(z - (1 + \alpha)T(z))$ with $\tilde{g}$ decreasing and where α is a constant parameter delivers an optimal tax with a constant marginal tax rate $T'(z) = 1/(1 + \alpha)$ and is empirically calibrated in online Appendix C2.

B. Transfers and Freeloaders

The public policy debate on transfers often focuses on whether nonworkers are deserving of support or not. Transfer beneficiaries are deemed deserving if they are truly unable to work, that is, if absent any transfers, they would still not work and live in great poverty without resources. Conversely, they are considered non-deserving, or “freeloaders” if they could work and would do so absent more generous transfers. The presence of such “freeloaders,” perceived to take undue advantage of a generous transfer system, is precisely why many oppose welfare (see, e.g., Ellwood 1988). It is also the reason why many welfare programs try to target populations which are deemed more vulnerable and less prone to taking advantage of the system. Historically, disabled people, widows, and later on single parents have been most likely to receive public transfers.²¹ Our online survey in online Appendix C confirms that people have strong views on who, among those out-of-work, is deserving of support. We consider a basic model to explain how generalized weights can be used to capture the concept of freeloaders.

Model.—Starting from our general model, assume that individuals can either work and earn $z = 1$, or not work and earn zero $z = 0$. We take the special

²¹ The origins of the US welfare system since 1935, starting with the Aid to Families with Dependent Children and continuing with the Temporary Assistance to Needy Families assistance programs highlights that logic.

functional utility form: $u(c - \theta \cdot z)$ where $z \in \{0, 1\}$. Consumption c is equal to c_0 if an individual is out of work and to $c_1 = (1 - \tau) + c_0$ if she works, so that τ is the net tax rate on earnings.²² Taxes fund the transfer c_0 . The cost of work θ is private information and distributed according to a cdf $P(\theta)$. An individual with cost of work θ works if and only if $\theta \leq c_1 - c_0 = 1 - \tau$. Hence, the fraction of people working is $P(1 - \tau)$. Let e be the elasticity of total earnings $P(1 - \tau)$ with respect to $1 - \tau$.

This model is a special case of the optimal linear tax model discussed in Section I. Hence, we can immediately apply formula (3) and the optimal tax is $\tau^* = (1 - \bar{g})/(1 - \bar{g} + e)$. As nonworkers have $z_i = 0$ and workers have $z_i = 1$, we have $\bar{g} = \int_i g_i z_i di / (\int_i g_i di \cdot \int_i z_i di) = \bar{g}_1 / [P \cdot \bar{g}_1 + (1 - P) \cdot \bar{g}_0]$ where $P = P(1 - \tau)$, $\bar{g}_1$ is the average g_i on workers, and $\bar{g}_0$ is the average g_i on nonworkers.

Standard Utilitarian Approach.—Under the utilitarian objective, we have $g_i = u'(c_0)$ for all nonworkers so that $\bar{g}_0 = u'(c_0)$.²³ Hence, the utilitarian approach cannot distinguish between the deserving poor and freeloaders. The utilitarian social welfare weight placed on nonworkers depends only on c_0 and is completely independent of whether the person would have worked absent taxes and transfers: The standard approach cannot take into account counterfactuals.

Generalized Social Welfare Weights.—The generalized weights allow us to treat the deserving poor differently from the freeloaders. The deserving poor are those with $\theta > 1$, (who would not work, even absent any transfer). The freeloaders are those with $1 \geq \theta > 1 - \tau$ (who do not work because of the existence of transfers). Denoting by $P_0 = P(1)$ the fraction working when $\tau = 0$, there are $P = P(1 - \tau)$ workers, $1 - P_0$ deserving poor, and $P_0 - P$ freeloaders.

Let us assume that society sets social marginal welfare weights as follows. Workers obtain a standard weight $g_i = u'(c_1 - \theta_i)$ if $z_i = 1$. The deserving poor obtain a standard weight $g_i = u'(c_0)$ if $z_i = 0$ and $\theta_i \geq 1$. Finally, the freeloaders obtain a weight $g_i = 0$ if $z_i = 0$ and $\theta_i < 1$. The weight function is $g_i = g(c_i, z_i; \theta_i, \tau)$. In our general notation, $x_i^b = (\theta_i, \tau)$.²⁴

With such weights, we have $\bar{g}_0 = u'(c_0) \cdot (1 - P_0)/(1 - P)$, as only a fraction $(1 - P_0)/(1 - P)$ of the nonworkers are deserving. Hence, $\bar{g}_0$ is lower relative to the utilitarian case. $\bar{g}_1$ is unchanged relative to the utilitarian case. Therefore, $\bar{g} = \bar{g}_1 / [P \cdot \bar{g}_1 + (1 - P) \cdot \bar{g}_0]$ is now *higher* than in the utilitarian case and τ^* is correspondingly lower (keeping e and P constant). As expected, the presence of freeloaders reduces the optimal tax rate τ^* relative to the standard case.

In the extreme case in which all nonworkers are freeloaders, the optimal transfer (and hence the tax rate financing it) is zero. This corresponds to the (extreme) view that all nonemployment is created by an overly generous transfer system. As long as there are some deserving poor though, the tax rate and transfers will be positive.

²² As there are only two earnings outcomes, 0 and 1, this tax system is fully general.

²³ For workers, $g_i = u'(c_1 - \theta_i)$ so that $\bar{g}_1 = \int_{\theta \leq 1 - \tau} u'(c_0 + (1 - \tau) - \theta) dP(\theta)/P < \bar{g}_0$.

²⁴ Note that the cost of work θ_i enters into the social welfare weights so that it is fair to compensate people for their differential cost of work. If instead θ reflected pure laziness, then this might not be the case (as in the Fair Income Tax theory of Fleurbaey and Maniquet 2011, also covered in online Appendix B5).

For a given τ , when e is larger, $(1 - P_0)/(1 - P)$ is smaller as more people become freeloaders. Hence a higher e reduces τ^* not only through the standard efficiency effect e but also through the channel $\bar{g}$ as it negatively affects society's view on how deserving the poor are.

This example also illustrates that ex post, it is possible to find suitable Pareto weights for the welfarist approach that can rationalize the tax rate τ^* obtained with generalized welfare weights. In this case, maximizing $\int_{\theta} \omega(\theta) \cdot u(c - \theta \cdot z) \cdot dP(\theta)$ with $\omega(\theta) = 1$ for $\theta \leq 1 - \tau^*$ (workers) and $\theta \geq 1$ (deserving poor) and $\omega(\theta) = 0$ for $1 - \tau^* < \theta < 1$ (freeloaders). However, the Pareto weights ω depend on the optimum tax rate τ^* and, hence, cannot be specified ex ante. Note that our approach is not equivalent to making the Pareto weights endogenous. Indeed, a social welfare function of the form: $\int_{\theta} \omega(\theta, \tau) \cdot u(c - \theta \cdot z) \cdot dP(\theta)$ with $\omega(\theta, \tau) = 1$ for $\theta \leq 1 - \tau$ and $\theta \geq 1$ and $\omega(\theta, \tau) = 0$ for $1 - \tau < \theta < 1$ would not yield the same solution because we would have to take into account the change in social welfare stemming from changes in the endogenous Pareto weights. What is more, such a social welfare objective with endogenous Pareto weights could yield Pareto dominated outcomes.²⁵ Instead, our approach consists in weighting utility changes using locally fixed weights, which respects the local Pareto principle.

APPLICATION 1 (Desirability of In-Work Benefits): *As shown in Piketty and Saez (2013a), in the nonlinear tax model of Section I, when some individuals do not work, the optimal marginal tax rate at the bottom with zero earnings takes the simple form $T'(0) = (\bar{g}_0 - 1)/(\bar{g}_0 - 1 + e_0)$ where $\bar{g}_0$ is the average social marginal welfare weight on those out-of-work and e_0 is (minus) the elasticity of the fraction of individuals out-of-work with respect to $1 - T'(0)$. Unlike in the standard utilitarian case, in which $\bar{g}_0 > 1$ and hence $T'(0) > 0$, with freeloaders, $\bar{g}_0$ is lower and could be lower than one, in which case $T'(0) < 0$, i.e., in-work benefits are optimal.*

APPLICATION 2 (Transfers over the Business Cycle): *Individuals are less likely to be responsible for their unemployment status in a recession than in an expansion, so that the composition of those out of work changes over the business cycle. Hence, this is a force pushing toward expanding benefits in bad times. Results from the online survey show indeed that support for the unemployed depends strongly on whether they can or cannot find jobs (see online Appendix C).*

C. Tagging and Horizontal Equity Concerns

Under welfarism, if agents can be separated into different groups, based on attributes, so-called “tags,” which are correlated with earnings ability or behavioral elasticities, then an optimal tax system should generally have differentiated taxes across those groups.²⁶ Mankiw and Weinzierl (2010) explore a tax schedule differentiated by height and use this stark example as a critique of utilitarianism. In practice,

²⁵ Indeed, reforms that increase the weights could be desirable even if all agents' utilities decline slightly.

²⁶ Some attributes can be perfect tags in the sense of being impossible to influence by the agent. An example would be height, which has been shown to be positively correlated with earnings (see Mankiw and Weinzierl 2010), or gender. Others are potentially elastic to taxes (such as the number of children or marital status).

society seems to oppose taxation based on such characteristics, probably because it is deemed unfair to tax differently people with the same ability to pay. These “horizontal equity” concerns, or the wish to treat “equals as equals” seem important in practice and a realistic framework for optimal tax policy needs to be able to include them.²⁷

It is possible to capture horizontal equity concerns using generalized social welfare weights if we extend our basic framework to allow the weights to be dependent on the *reform considered* (instead of depending solely on the current level of taxes and transfers as we have done so far). To save space, we will not develop a general theory but will solely focus on a simple example.

To illustrate this in the simplest way possible, we consider a Ramsey tax problem where linear taxation is designed to raise a given (non-transfer) revenue E . There are two groups which differ according to an observable and perfectly inelastic attribute $m \in \{1, 2\}$. The two groups differ according to their taxable income elasticities (respectively denoted e_1 and e_2). The tax rate on group m is denoted by τ_m . The budget constraint, with multiplier p , is

$$(5) \quad \tau_1 \cdot \int_{i \in 1} z_i \, di + \tau_2 \cdot \int_{i \in 2} z_i \, di \geq E.$$

We abstract entirely from vertical equity issues by assuming that utilities are linear in consumption so that $u_{c_i} = 1$. As a result, under the utilitarian criterion, all individuals have the same social marginal welfare weight. The standard utilitarian approach would naturally lead to different tax rates for each group, following standard Ramsey considerations, such that $\tau_m = (1 - 1/p)/(1 - 1/p + e_m)$ where $p > 0$ is determined so that the tax system raises exactly E to meet the budget constraint (5) (see online Appendix A5 for a formal proof). Hence, the standard tax system would generate horizontal inequities, i.e., some individuals are taxed more than others based on a tag and conditional on income. Without loss of generality, we assume $e_2 > e_1$ so that group 2 is more elastic and should optimally be taxed less.

How can we specify generalized social marginal welfare weights to reflect the view that horizontal inequities are unfair? First, as in the utilitarian case, absent any horizontal inequity (i.e., if $\tau_1 = \tau_2$), we assume that weights are equal across the population (and normalized to one). Second, if the tax system creates horizontal inequity (i.e., $\tau_1 \neq \tau_2$), then social marginal welfare weights are equal to one for individuals in the group facing the highest tax rate and zero for those facing the lowest tax rate. However, such weights are not sufficient to ensure that the no tagging tax system with $\tau_1 = \tau_2$ is a local optimum. Indeed, starting from this tax system, lowering τ_2 and increasing τ_1 would be desirable as everybody has the same weights (no equilibrium would exist). Hence and last, we need to expand our framework and make weights depend on the direction of reform as well when we start from an equitable tax system and the small reform introduces a horizontal inequity.²⁸

²⁷ Kaplow (2001) criticizes the concept of horizontal equity and highlights that it will conflict with the Pareto principle in some cases. Our nonnegative social welfare weights guarantee that this cannot occur in our setup.

²⁸ If the system is inequitable to start with (i.e., $\tau_1 \neq \tau_2$), then a small reform will not affect the sign of the initial inequity and hence the weights do not need to depend on the reform.

In our simple example with linear taxation, starting from identical tax rates for the two groups, a small reform that introduces inequity can lead to (i) $\tau_1 > \tau_2$, or (ii) $\tau_1 < \tau_2$. In case (i), group 1 suffers from a horizontal inequity because of the reform and the weights are set equal to one on group 1 and zero on group 2. In case (ii) conversely, the weights are set equal to zero on group 1 and one on group 2. Such weights are designed to avoid horizontal inequities unless they benefit the group discriminated against.

Regularity Assumptions.—We assume that there is a uniform tax rate $\tau_1 = \tau_2 = \tau^*$ that can raise E . We assume that the tax functions $\tau_1 \rightarrow \tau_1 \cdot \int_{i \in 1} z_i di$ and $\tau_2 \rightarrow \tau_2 \cdot \int_{i \in 2} z_i di$, and the uniform rate tax function $\tau \rightarrow \tau \cdot (\int_{i \in 1} z_i di + \int_{i \in 2} z_i di)$ are all single peaked (Laffer curves).

Naturally, the peaks are, respectively, at $\tau_1 = 1/(1 + e_1)$, $\tau_2 = 1/(1 + e_2)$, and $\tau = 1/(1 + e)$, where e is the elasticity of total income with respect to $1 - \tau$.

PROPOSITION 6: *Let τ^* be the (smallest) uniform rate that raises E : $\tau^* (\int_{i \in 1} z_i di + \int_{i \in 2} z_i di) = E$.*

- If $1/(1 + e_2) \geq \tau^*$ then the only local optimum has horizontal equity with $\tau_1 = \tau_2 = \tau^*$.
- If $1/(1 + e_2) < \tau^*$ then the only local optimum has horizontal inequity with $\tau_2^* = 1/(1 + e_2) < \tau^*$ (revenue maximizing rate on group 2) and $\tau_1^* < \tau^*$ is the smallest τ_1 such that $\tau_1 \cdot \int_{i \in 1} z_i di + \tau_2^* \cdot \int_{i \in 2} z_i di = E$. This optimum Pareto dominates the uniform tax system $\tau_1 = \tau_2 = \tau^*$.

Therefore, a tax system with horizontal inequity can be an optimum only if it helps the group discriminated against, i.e., no reform can improve the welfare of those discriminated against. This can happen only when tagging creates a Pareto improvement, which dramatically reduces the scope for using tagging in practice. In our example, if the government wants to set τ_2 at a lower level than τ_1 , then τ_2 must necessarily be set at the revenue maximizing rate.

This is reminiscent of a Rawlsian setup, in which society only cares about the least well-off. Here, the set of people whom society cares about is endogenous to the tax system. Namely, they are the ones discriminated against because of tagging. In other words, we can rephrase the Rawlsian criterion as follows: it is permissible to discriminate against a group using taxes and transfers not based on ability to pay only in the case where such discrimination allows to improve the welfare of the group discriminated against.

III. Link with Alternative Justice Principles

In this section, we illustrate how our framework can be connected to justice principles that are not captured by the standard welfarist approach but have been discussed in the normative tax policy literature. We use formula (2) to show how social welfare weights derived from alternative justice principles map into optimal tax formulas.

A. Equality of Opportunity

To capture the concept of equality of opportunity, Roemer (1998) and Roemer et al. (2003) consider models where individuals differ in their ability to earn, but part of the ability is due to family background (which individuals are not responsible for) and to merit (which individuals are responsible for). Conditional on family background, merit is measured by the percentile of the earnings distribution the individual is in. Society is willing to redistribute across family backgrounds but not across earnings conditional on family background.

Formally, individual i 's utility is $u_i = u(c_i - v(z_i/w_i))$. w_i is productivity stemming from two sources. First, individual i is born into a low or high family background denoted by $B_i = 0, 1$. Second, through merit (e.g., hard work at school), individual i reaches rank r_i conditional on background and hence obtains a productivity w_i at the r_i th percentile of the distribution of productivity conditional on B_i , $F(w|B_i)$. For any rank r , productivity is higher when coming from the high background.

Once w_i is realized, individuals have identical utility (and hence earnings behavior) regardless of background. Let $\bar{c}(r) \equiv (\int_{(i: r_i=r)} c_i di) / \text{Prob}(i : r_i = r)$ be the average consumption of those at conditional rank r . Equality of opportunity can be captured by weights of the form $g_i = g(c_i; \bar{c}(r_i)) = 1(c_i \leq \bar{c}(r_i))$, with $x_i^s = \bar{c}(r_i)$, $x_i^u = w_i$ and x_i^b empty. Weights are concentrated on those who consume less than average conditional on within background rank (i.e., merit).

We assume that the government cannot observe family background and hence has to base taxes and transfers on earnings z only.²⁹ In that case, with any $T(z)$ such that $T'(z) < 1$, conditional on rank r ; individuals from an advantaged background earn more and consume more than those from a disadvantaged background. Hence, $g_i = 1$ for all individuals from the disadvantaged background and $g_i = 0$ for all individuals from the advantaged background. Next, we aggregate the weights at each z level. They imply that $\bar{G}(z)$ is the fraction of individuals from a disadvantaged background earning at least z relative to the population wide fraction of individuals from a disadvantaged background. This is also known as the representation index. This leads to the social welfare function proposed by Roemer et al. (2003) which counts only the utility of those from a disadvantaged background. Naturally, we expect $\bar{G}(z)$ to decrease with z as it is harder for those coming from a disadvantaged background to reach upper incomes. If the representation of individuals from a disadvantaged background is zero at the top, the top tax rate should be set to maximize tax revenue. Hence, equality of opportunity provides a justification for having social welfare weights decreasing with income, which is orthogonal to the utilitarian mechanism of decreasing marginal utility of consumption.

Calibrating the Weights to US Intergenerational Mobility.—The US intergenerational income mobility statistics produced by Chetty et al. (2014) can be used to

²⁹In reality, family background is observable, but the real advantage derived from family background for each individual is not. As a result, family background is an imperfect tag for ability advantage derived from family background. Society could be reluctant to use such a tag because of horizontal equity concerns as discussed above. A person might come from a privileged family background and yet might not have received much parental support. Taxing such a person more (conditional on earnings) would be unfair.

illustrate this discussion. Suppose we define low background as having parents coming from the bottom half of the income distribution of parents. Column 1 in Table 2 displays the fraction of individuals with parents below median income above various percentiles of the income distribution.³⁰ As individuals with parents below median are by definition half of the population, $\bar{G}(z)$ is simply half of column 1 and is reported in column 2. $\bar{G}(z)$ falls from 1 at percentile 0 (by definition) to 0.333 at the 99.9th percentile. Hence, in contrast to the standard utilitarian case where $\bar{G}(z)$ converges to zero for large z (with a concave utility function with marginal utility converging to zero), in the equality of opportunity case, $\bar{G}(z)$ converges to a positive value of one-third because a substantial fraction of high earners come from a disadvantaged background. $\bar{G}(z)$ appears stable within the 99th percentile as $\bar{G}(z)$ is virtually the same at the 99th percentile and the 99.9th percentile. Hence, under this equality of opportunity criterion, individuals at the 99.9th percentile are deemed no less deserving than individuals at the lower 99th percentile because they are equally likely to come from a disadvantaged background.

This has two important optimal tax consequences for top earners. First, with a Pareto parameter $a = 1.5$ and an elasticity $e = 0.5$, the optimal top asymptotic tax rate is $\tau = 1/(1 + a \cdot e) = 0.57$ in the utilitarian case and $\tau = (1 - 1/3)/(1 - 1/3 + a \cdot e) = 0.47$ in the equality of opportunity case.³¹ Second, a society which values individuals coming from a low background would use progressive income taxation but the top tax rate would be stable within the top 1 percent because the representation of individuals from a disadvantaged background is stable above the ninety-ninth percentile.

To illustrate these properties, Table 2, column 3 presents optimal marginal tax rates at various income levels using formula (2). The weights $\bar{G}(z)$ are from column 2. We calibrate $\alpha(z)$ using the actual distribution of income based on 2008 income tax return data, the latest year available. We use a constant elasticity $e = 0.5$ which is a mid to upper range estimate based on the literature (see Saez, Slemrod, and Gieritz 2012). Because of uncertainty in the level of e , the simulations should be considered as illustrative at best. Column 3 shows that optimal marginal tax rates are U-shaped but about constant above the ninety-ninth percentile. For comparison, columns 4 and 5 present the utilitarian weights $\bar{G}(z)$ and optimal marginal tax rates $T'(z)$ assuming a log-utility so that the welfare weight $\bar{g}(z)$ at income level z , is proportional to $1/(z - T(z))$. The utilitarian case delivers optimal tax rates that are about ten points higher than the equality of opportunity case and significantly more progressive.

B. Poverty Alleviation

The poverty rate, defined as the fraction of households below a given disposable income threshold (the poverty threshold) attracts substantial attention in the public debate. Hence, it is conceivable that governments aim to either reduce the poverty

³⁰ These estimates are based on all US individuals born in 1980–1981 with their income measured at age 30–31. In this simulation, we take a shortcut and assume these estimates hold more broadly in the full population.

³¹ Naturally, in a less meritocratic society than the United States at present, $\bar{G}(z)$ for large z could possibly be smaller and the optimal top tax rate correspondingly closer to the optimal utilitarian tax rate.

TABLE 2—EQUALITY OF OPPORTUNITY VERSUS UTILITARIAN OPTIMAL TAX RATES

	Equality of opportunity			Utilitarian (log-utility)	
	Fraction from low background (= parents below median) above each percentile (1)	Implied social welfare weight $\bar{G}(z)$ above each percentile (2)	Implied optimal marginal tax rate at each percentile (in percent) (3)	Utilitarian social welfare weight $\bar{G}(z)$ above each percentile (4)	Utilitarian optimal marginal tax rate at each percentile (in percent) (5)
<i>Income percentile</i>					
$z = 25\text{th percentile}$	0.443	0.886	53	0.793	67
$z = 50\text{th percentile}$	0.373	0.746	45	0.574	58
$z = 75\text{th percentile}$	0.303	0.606	40	0.385	51
$z = 90\text{th percentile}$	0.236	0.472	34	0.255	42
$z = 99\text{th percentile}$	0.170	0.340	46	0.077	54
$z = 99.9\text{th percentile}$	0.165	0.330	47	0.016	56

Notes: This table compares optimal marginal tax rates at various percentiles of the distribution (listed by row) using an equality of opportunity criterion (in column 3) and a standard utilitarian criterion (in column 5). Both columns use the optimal tax formula $T'(z) = [1 - \bar{G}(z)]/[1 - \bar{G}(z) + \alpha(z) \cdot e]$ discussed in the text where $\bar{G}(z)$ is the average social marginal welfare weight above income level z , $\alpha(z) = (zh(z))/(1 - H(z))$ is the local Pareto parameter (with $h(z)$ the density of income at z , and $H(z)$ the cumulative distribution), and e the elasticity of reported income with respect to $1 - T'(z)$. We assume $e = 0.5$. We calibrate $\alpha(z)$ using the actual distribution of income based on 2008 income tax return data (and ignoring the effects of changing taxes on $\alpha(z)$). For the equality of opportunity criterion, $\bar{G}(z)$ is the representation index of individuals with income above z who come from a disadvantaged background (defined as having a parent with income below the median). This representation index is estimated using the national intergenerational mobility statistics of Chetty et al. (2014) based on all US individuals born in 1980–1981 with their income measured at age 30–31. For the utilitarian criterion, we assume a log-utility so that the social welfare weight $\bar{g}(z)$ at income level z is proportional to $1/(z - T(z))$.

gap (defined as the amount of money needed to lift all households out of poverty) or reduce the poverty rate (the number of households below the poverty threshold). A few studies have considered government objectives incorporating such poverty concerns. Besley and Coate (1992) and Kanbur, Keen, and Tuomala (1994) show how adopting poverty minimization indexes affects optimal tax analysis. Importantly, their findings imply that the outcomes can be Pareto dominated. In this section, we show how generalized welfare weights can incorporate poverty alleviation considerations in the traditional optimal tax analysis while maintaining the Pareto principle.

Let us denote the poverty threshold by $\bar{c}$. Anybody with disposable income $c < \bar{c}$ is poor. Utility is taken to be a special case of our general formulation: $u_i = u(c_i - v(z_i/w_i))$.

Poverty Gap Minimization and Implicit Negative Weights.—Kanbur, Keen, and Tuomala (1994) derive the tax system minimizing the poverty gap. If the lowest ability agent exerts positive labor supply, the authors find that the bottom marginal tax rate should be negative as illustrated in Figure 2, panel A in a (pretax income, posttax income) plane. It is well-known, however, that in the welfarist case, the optimal tax rate at the bottom cannot be negative if everybody works and the lowest earnings are strictly positive (Seade 1977). Indeed, starting from a negative bottom rate at minimum earnings level $z_b > 0$, slightly increasing the bottom marginal tax rate below z_b would be Pareto improving without violating incentive constraints: it allows the lowest productivity agent to work less, which is welfare improving, and

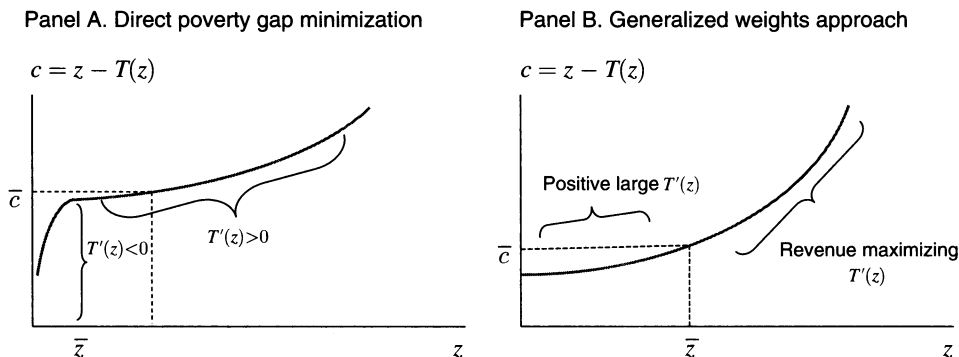


FIGURE 2. OPTIMAL POLICIES FOR POVERTY GAP MINIMIZATION

Notes: The figure displays the optimal tax schedule for poverty gap alleviation in a (pretax income z , posttax income $c = z - T(z)$) plane. Panel A plots the schedule for the approach that consists of directly minimizing the poverty gap (Kanbur, Keen, and Tuomala 1994). The marginal tax rate is negative below the poverty threshold $\bar{z}$. Panel B plots the schedule derived using generalized welfare weights concentrated on those in poverty. The optimal tax schedule is similar to the standard utilitarian case with high marginal tax rates at the bottom.

also raises more revenue, as the marginal tax rate is negative to start with. This is a Pareto improvement.³²

The discrepancy between the poverty gap minimization and the welfarist objective arises because the former does not take into account the disutility from work for the lowest productivity agents. Pushing them to work more is always desirable under a poverty gap objective.

Generalized Social Welfare Weights.—Let us instead consider our generalized weights approach. If the demogrant can be made bigger than $\bar{c}$, then the optimum way to fight poverty is to raise enough taxes to set the demogrant equal to $\bar{c}$. Once the poverty threshold has been attained, there is no reason to have differences in social welfare weights and hence the weights would all be equal to a fixed g for those with positive earnings so that $\bar{G}(z) = g$ for $z > 0$. Using formula (2), we would have $T'(z) = (1 - g)/(1 - g + \alpha(z) \cdot e(z))$, where g is set so that total taxes collected raise enough revenue to fund the demogrant $\bar{c}$. The less trivial case is when even with $g = 0$, tax revenue cannot fund a demogrant as large as $\bar{c}$. Let us denote by $\bar{z}$ the (endogenous) earnings level such that $\bar{c} = \bar{z} - T(\bar{z})$, i.e., that defines the pretax poverty threshold.

A natural way to capture a “poverty gap alleviation” objective is to assume that social welfare weights are concentrated among individuals with disposable income c below the poverty threshold $\bar{c}$. We can therefore specify the generalized welfare weights as follows: $g(c, z; \bar{c}) = 1$ if $c < \bar{c}$ and $g(c, z; \bar{c}) = 0$ if $c \geq \bar{c}$.³³ In this case, $x_i^u = w_i$, $x_i^s = \bar{c}$, and x^b is empty. We have $\bar{g}(z) = 0$ for $z \geq \bar{z}$ and $\bar{g}(z) = 1/H(\bar{z})$ for $z < \bar{z}$ so that $\int_0^\infty \bar{g}(z) dH(z) = 1$. Hence, we have $\bar{G}(z) = 0$ for $z \geq \bar{z}$. We obtain $\bar{G}(z) = [1 - H(z)/H(\bar{z})]/[1 - H(z)]$ for $z < \bar{z}$.

³²The proof is exactly symmetrical to the proof of the famous zero marginal top tax rate result.

³³A less extreme (but still tractable) version of this assumption would set $g(c, z; \bar{c}) = \underline{g}$ if $c \geq \bar{c}$ with $0 < \underline{g} < 1$.

(as $\bar{G}(z)[1 - H(z)] = \int_z^{\bar{z}} \bar{g}(z') dH(z')$). Applying formula (2) yields the following proposition.

PROPOSITION 7: *The optimal tax schedule that minimizes the poverty gap is*

$$T'(z) = \frac{1}{1 + \alpha(z) \cdot e(z)} \quad \text{if } z > \bar{z}$$

$$T'(z) = \frac{[1/H(\bar{z}) - 1]H(z)}{[1/H(\bar{z}) - 1]H(z) + \alpha(z)[1 - H(z)] \cdot e(z)} \quad \text{if } z \leq \bar{z}.$$

As $[1/H(\bar{z}) - 1]H(\bar{z}) = 1 - H(\bar{z})$, the marginal tax rate is continuous at the poverty threshold $\bar{z}$. The marginal tax rate maximizes revenue above $\bar{z}$ and is positive (and typically large) below $\bar{z}$. The shape of optimal tax rates is quite similar to the standard utilitarian case and is illustrated in Figure 2, panel B in a (pretax income, posttax income) plane. Our approach can be viewed as minimizing the poverty gap while respecting the Pareto principle.³⁴

IV. Conclusion

This paper has shown that the concept of *generalized marginal social welfare weights* is a fruitful way to extend the standard welfarist theory of optimal taxation. The use of suitable generalized social welfare weights can help enrich the traditional welfarist approach and account for existing tax policy debates and structures while retaining (local) Pareto constrained efficiency. Our theory brings back social preferences as a critical element for optimal tax theory analysis. Naturally, this flexibility of generalized social weights begs the question of what social welfare weights ought to be and how they are formed.

Generalized welfare weights can be derived from social justice principles, leading to a normative theory of taxation. The most famous example is the Rawlsian theory where the generalized social marginal welfare weights are concentrated solely on the most disadvantaged members of society. As we have discussed, binary weights (equal to one for those deserving more support and zero otherwise) have normative appeal and can be used in a broad range of cases. The Rawlsian case can also be extended to a discrete number of groups, ranked according to deservedness, such that society has redistributive preferences across groups but libertarian preferences within groups. Naturally, who is deserving might itself be endogenous to the tax system. Such weights can also prioritize justice principles in a lexicographic form.

First, injustices created by tax policy (such as horizontal inequities) may have the highest priority. In that case, those deserving of support are those discriminated against whenever horizontal inequities arise. Hence, horizontal inequities are allowed only if they help the group discriminated against, dramatically lowering the scope for such policies (such as tagging) that are recommended by the standard welfarist approach but typically not observed in practice.

³⁴ “Poverty rate minimization,” where the government attempts to minimize the number of people living below the poverty line by concentrating weights on those at the poverty threshold, is treated in online Appendix B4.

Second, deserving individuals will be those who face difficult economic situations through no fault of their own. This captures the principle of compensation. Health expenses come to mind, explaining why virtually all advanced countries adopt generous public health insurance that effectively compensate individuals for the bad luck of facing high health expenses. Once disparities in health care costs have been compensated by public health insurance provision, this element naturally drops out of social welfare weights. Family background is obviously another element that affects outcomes and that individuals do not choose. This explains why equality of opportunity has wide normative appeal both among liberals and conservatives. Policies aiming directly to curb such inequities such as public education or inheritance taxation can therefore be justified on such grounds.³⁵ Naturally, public education or inheritance taxation cannot fully erase inequalities due to family background. This leaves a role for taxes and transfers based on income that aim at correcting remaining inequities in opportunity as in the theory of Roemer et al. (1993) which can be implemented using intergenerational mobility statistics.

Third, even conditional on background, there remains substantial inequality in incomes. Part of this inequality is due to choices (preferences for leisure versus consumption) but part is due to luck (ability and temperament are often not based on choice). Naturally, there is a debate on the relative importance of choices versus luck, which impacts the resulting social welfare weights.³⁶ As in the fair income tax theory, the generalized social welfare weights have the advantage of highlighting which differences society considers unfair (for example, due to intrinsic skill differences) and which it considers fair (for example, due to different preferences for work).³⁷

Finally, there might be scope for redistribution based on more standard utilitarian principles, i.e., the fact that an additional dollar of consumption matters more for lower income individuals than for higher income individuals. In the public debate, this principle seems relevant at the low income end to justify the use of anti-poverty programs but less widely invoked to justify progressive taxation at the upper end.

Social preferences are indeed shaped by beliefs about what drives disparities in individual economic outcomes (effort, luck, background, etc.) as in the model of Piketty (1995). We show in online Appendix C that online surveys can be used to empirically estimate social preferences, leading to a positive theory of taxation. Alternatively, the inverse optimum tax literature estimates the weights that would make the current tax system optimal. More ambitiously, economists can also cast light on those mechanisms and hence enlighten the public perceptions so as to move the debate up to the higher level of normative principles.

³⁵ Piketty and Saez (2013b) consider optimal inheritance taxation with such “meritocratic” weights concentrated on those receiving no inheritance.

³⁶ The case of luck versus deserved income is treated in online Appendix B2.

³⁷ See the treatment of the fair income tax theory in online Appendix B5.

REFERENCES

- Alesina, Alberto, and George-Marios Angeletos. 2005. "Fairness and Redistribution." *American Economic Review* 95 (4): 960–80.
- Besley, Timothy, and Stephen Coate. 1992. "Workfare versus Welfare: Incentive Arguments for Work Requirements in Poverty-Alleviation Programs." *American Economic Review* 82 (1): 249–61.
- Chetty, Raj. 2009. "Sufficient Statistics for Welfare Analysis: A Bridge Between Structural and Reduced-Form Methods." *Annual Review of Economics* 1: 451–88.
- Chetty, Raj, Nathaniel Hendren, Patrick Kline, and Emmanuel Saez. 2014. "Where is the Land of Opportunity? The Geography of Intergenerational Mobility in the United States." *Quarterly Journal of Economics* 129 (4): 1553–1623.
- Diamond, Peter A. 1998. "Optimal Income Taxation: An Example with a U-Shaped Pattern of Optimal Marginal Tax Rates." *American Economic Review* 88 (1) 83–95.
- Edgeworth, F. Y. 1897. "The Pure Theory of Taxation." *Economic Journal* 7 (25): 46–70.
- Ellwood, David T. 1988. *Poor Support: Poverty in the American Family*. New York: Basic Books.
- Fehr, Ernst, and Klaus M. Schmidt. 1999. "A Theory of Fairness, Competition, and Cooperation." *Quarterly Journal of Economics* 114 (3): 817–68.
- Fleurbaey, Marc, and François Maniquet. 2011. *A Theory of Fairness and Social Welfare*. Cambridge, UK: Cambridge University Press.
- Fleurbaey, Marc, and François Maniquet. 2015. "Optimal Taxation Theory and Principles of Fairness." CORE Discussion Paper 2015/5.
- Hendren, Nathaniel. 2013. "The Policy Elasticity." National Bureau of Economic Research Working Paper 19177.
- Jacquet, Laurence, and Etienne Lehmann. 2015. "Optimal Income Taxation when Skills and Behavioral Elasticities are Heterogeneous." CESifo Working Paper 5265.
- Kanbur, Ravi, Michael Keen, and Matti Tuomala. 1994. "Optimal Non-linear Income Taxation for the Alleviation of Income-Poverty." *European Economic Review* 38 (8): 1613–32.
- Kaplow, Louis. 2001. "Tax Policy and Horizontal Equity: Commentary." In *Inequality and Tax Policy*, edited by Kevin A. Hassett and R. Glenn Hubbard, 75–97. Washington, DC: American Enterprise Institute Press.
- Kleven, Henrik Jacobsen, and Claus Thustrup Kreiner. 2006. "The Marginal Cost of Public Funds: Hours of Work versus Labor Force Participation." *Journal of Public Economics* 90 (10–11): 1955–73.
- Mankiw, N. Gregory, and Matthew Weinzierl. 2010. "The Optimal Taxation of Height: A Case Study of Utilitarian Income Redistribution." *American Economic Journal: Economic Policy* 2 (1): 155–76.
- Mirrlees, J. A. 1971. "An Exploration in the Theory of Optimum Income Taxation." *Review of Economic Studies* 38 (2): 175–208.
- Piketty, Thomas. 1995. "Social Mobility and Redistributive Politics." *Quarterly Journal of Economics* 110 (3): 551–84.
- Piketty, Thomas, and Emmanuel Saez. 2013a. "Optimal Labor Income Taxation." In *Handbook of Public Economics*, Vol. 5, edited by Alan J. Auerbach, Raj Chetty, Martin Feldstein, and Emmanuel Saez, 391–474. Amsterdam: Elsevier Science B.V.
- Piketty, Thomas, and Emmanuel Saez. 2013b. "A Theory of Optimal Inheritance Taxation." *Econometrica* 81 (5): 1851–86.
- Roemer, John E. 1998. *Equality of Opportunity*. Cambridge, MA: Harvard University Press.
- Roemer, John E., Rolf Aaberge, Ugo Colombino, Johan Fritzell, Stephen P. Jenkins, Arnaud Lefranc, Ivo Marx et al. 2003. "To What Extent Do Fiscal Regimes Equalize Opportunities for Income Acquisition among Citizens?" *Journal of Public Economics* 87 (3–4): 539–65.
- Saez, Emmanuel. 2001. "Using Elasticities to Derive Optimal Income Tax Rates." *Review of Economic Studies* 68 (1): 205–229.
- Saez, Emmanuel, Joel Slemrod, and Seth H. Giertz. 2012. "The Elasticity of Taxable Income with Respect to Marginal Tax Rates: A Critical Review." *Journal of Economic Literature* 50 (1): 3–50.
- Saez, Emmanuel, and Stefanie Stantcheva. 2016. "Generalized Social Marginal Welfare Weights for Optimal Tax Theory: Dataset." *American Economic Review*. <http://dx.doi.org/10.1257/aer.20141362>.
- Seade, J. K. 1977. "On the Shape of Optimal Tax Schedules." *Journal of Public Economics* 7 (2): 203–35.
- Weinzierl, Matthew. 2014. "The Promise of Positive Optimal Taxation: Normative Diversity and a Role for Equal Sacrifice." *Journal of Public Economics* 118: 128–42.